

Autodesk® Scaleform®

AMP

Scaleform Analyzer for Memory and Performance AMP

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2013 4 10

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Autodesk® Scaleform® 4.4

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目次

1	はじめに	1
1.1	AMP	2
1.2	AMP	3
2	AMP の使い方	12
2.1		12
2.2		13
2.3		14
2.4		14
2.5		19
2.6		20
3	プラットフォームとインテグレーションに関する注意.....	22
4	ビルドに関する注意	23
5	AMP サポートの追加	24
5.1	Debug	24
5.2		24
5.3		25
5.4		25
6	AMP 関連の FAQ	26
7	追加情報	27

AMP™ Scaleform

CPU
Scaleform®

ActionScript

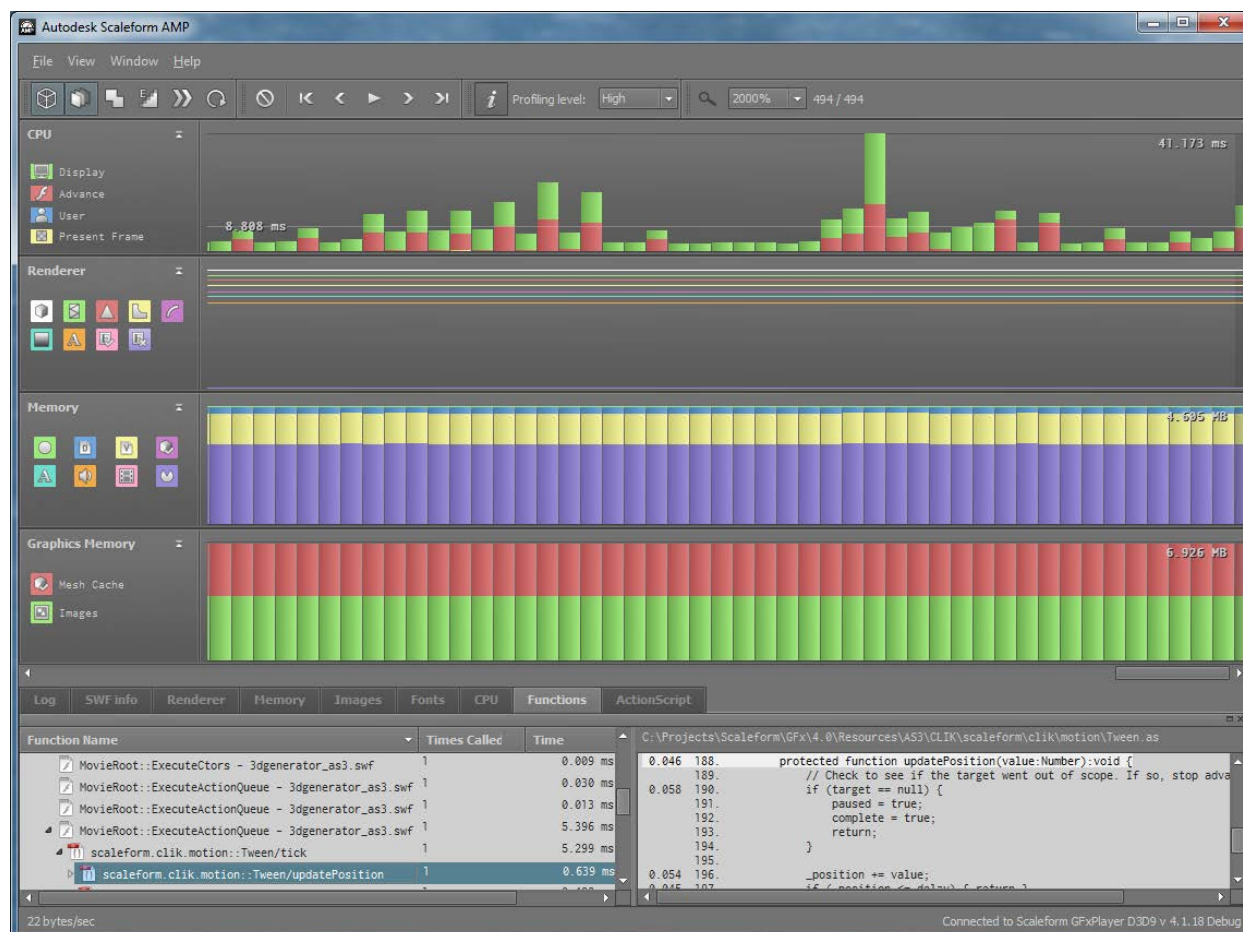


图 1: Scaleform AMP

Scaleform m

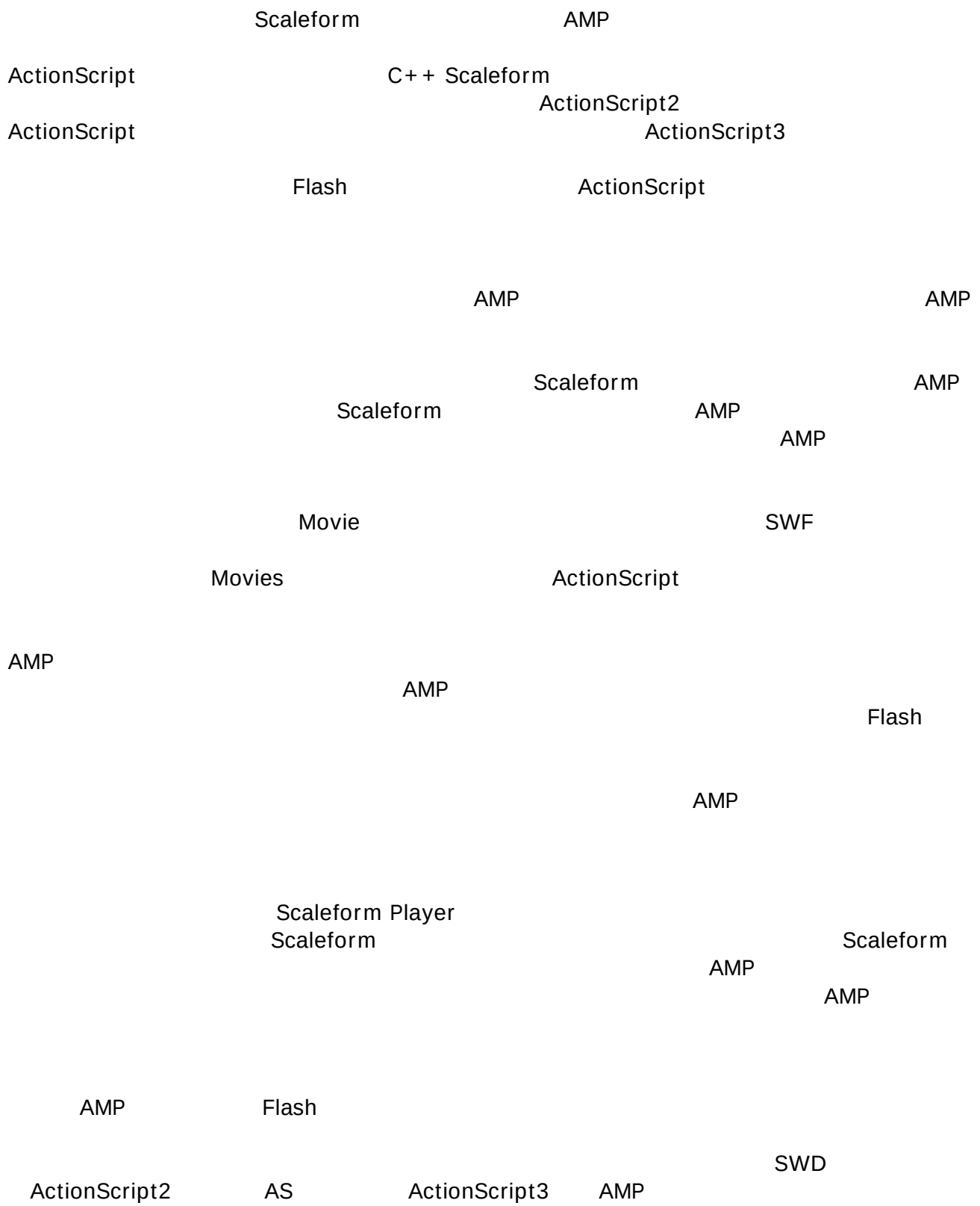
AMP

AMP

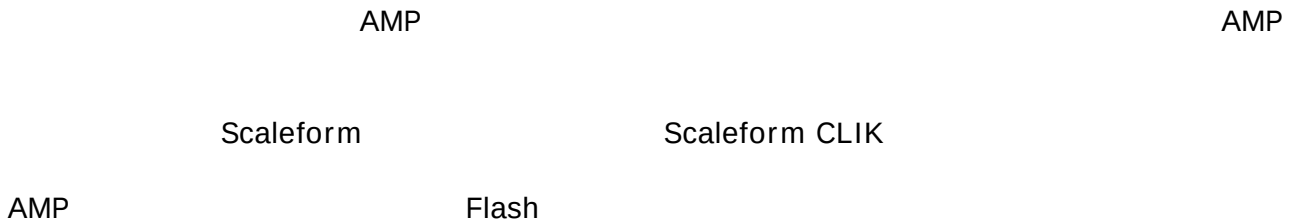
AMP

AMP Scaleform 3.2

1.1 AMP サーバー



1.2 AMP クライアント



- **CPU グラフ:** Scaleform (Display) Direct Access API (User) Scaleform CPU
(PresentFrame)

- **レンダリンググラフ:** [Scaleform Best Practices Guide](#) Scaleform

Scaleform

- **メモリーグラフ:** Scaleform

- **グラフィックスメモリーのグラフ:** Scaleform
AMP

GPU

- **Log タブ:** Log

Scaleform

- **Performance Adviser (パフォーマンスアドバイザー) :** Flash

- o *Text Fields optimized for readability:*

Anti-alias for readability

Anti-alias for animation

- o *Ratio of draw primitives to number of meshes*

1.0 1 1

- o *Number of draw primitives*

- o *Number of meshes*
- o *Number of triangles:*
- o *Mesh cache thrashing:*

- o *Number of strokes*
Flash
- o *Number of gradient fills*

- o *Number of masks*
- o *Number of filters*
- o *Number of DirectAccess API calls* API
- o *Number of rasterized glyphs*
- o *Number of font textures*
- o *Number of font texture updates* 1

- o *Font cache thrashing*

- o *Unused font texture percentage*

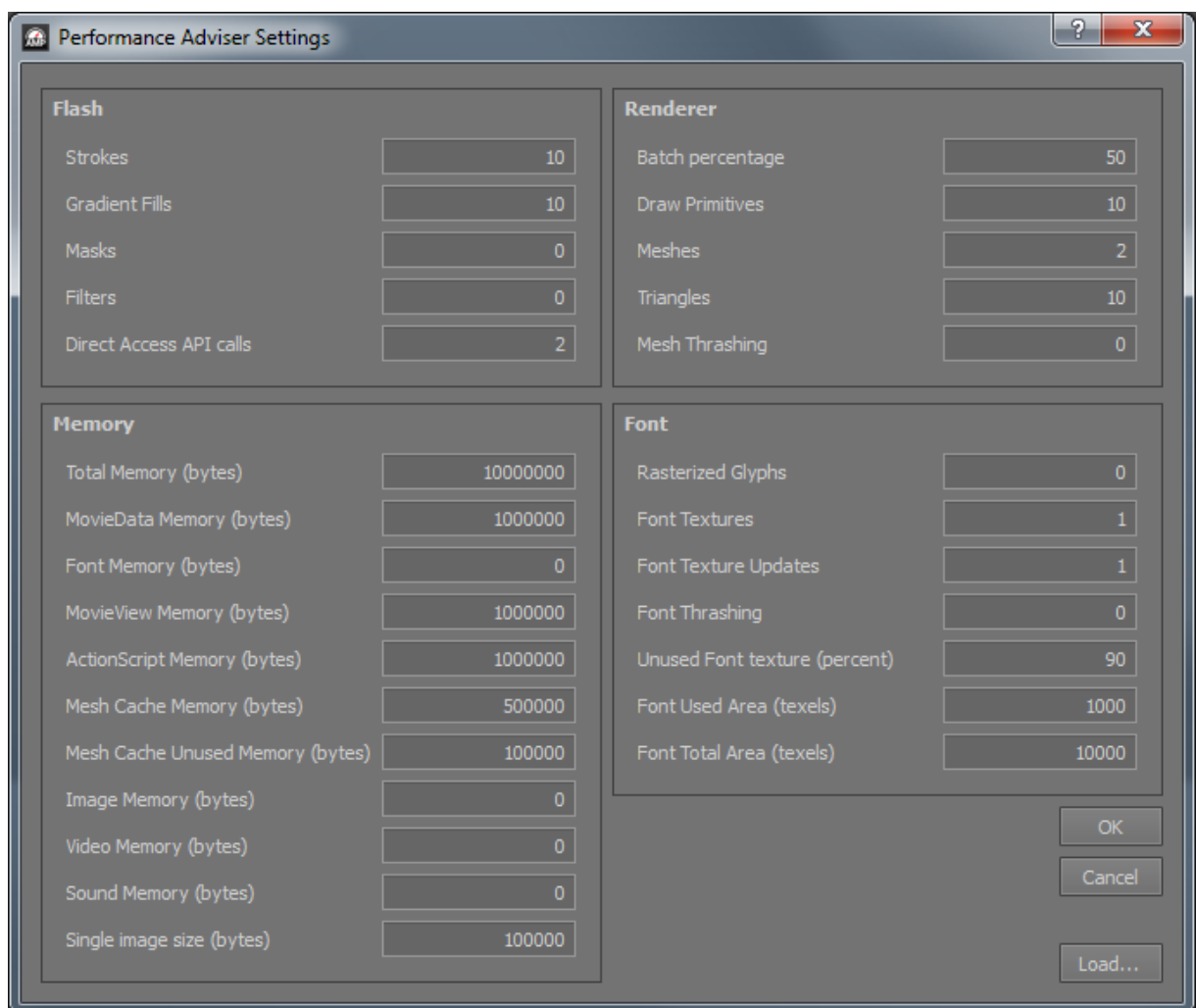
- o *Used font cache area*
- o *Total font cache area*

- o *Total Memory* Scaleform
- o *MovieData Memory*
- o *Font Memory*
- o *MovieView Memory* Gfx::Movie
MovieView
- o *ActionScript Memory* ActionScript
ActionScript

- o *Mesh Cache Memory*
- o *Unused Mesh Cache Memory*
- o *Image Memory*
- o *Video Memory* GPU
- o *Sound Memory*
- o *Number of images exceeding a certain size* AMP

adviser.xml XML AMP

Settings Performance Advisor



The image shows a screenshot of the 'Performance Adviser Settings' dialog box. The dialog is divided into four main sections: Flash, Memory, Renderer, and Font. Each section contains a list of settings with corresponding input fields.

Section	Setting	Value
Flash	Strokes	10
	Gradient Fills	10
	Masks	0
	Filters	0
	Direct Access API calls	2
Memory	Total Memory (bytes)	10000000
	MovieData Memory (bytes)	1000000
	Font Memory (bytes)	0
	MovieView Memory (bytes)	1000000
	ActionScript Memory (bytes)	1000000
	Mesh Cache Memory (bytes)	500000
	Mesh Cache Unused Memory (bytes)	100000
	Image Memory (bytes)	0
	Video Memory (bytes)	0
	Sound Memory (bytes)	0
Single image size (bytes)	100000	
Renderer	Batch percentage	50
	Draw Primitives	10
	Meshes	2
	Triangles	10
	Mesh Thrashing	0
Font	Rasterized Glyphs	0
	Font Textures	1
	Font Texture Updates	1
	Font Thrashing	0
	Unused Font texture (percent)	90
	Font Used Area (texels)	1000
	Font Total Area (texels)	10000

At the bottom right of the dialog, there are three buttons: 'OK', 'Cancel', and 'Load...'.

OK

XML

AMP

–adviser <AdviserFilePath>

AMP

-
- **SWF info タブ:**
- **Renderer タブ:**

- **Memory タブ:**

Scaleform

Scaleform

AMP

i

- **Images タブ:**

Scaleform

AMP

DXT

- **Fonts タブ:**

SWF

- **CPU タブ:**
User times

Advance

Display

- **Functions タブ:**

ActionScript C++
ActionScript VM

- **ActionScript タブ:**

Functions
C++

- **Source ウィンドウ:**
AS3
SWF

Functions
AS
SA3

ActionScript

AS2

High

AS2

Scaleform

2

Flash

AMP

AMP

SWF

AS3

SWF

- **File**

- o **Connection**

- o **Disconnect :**

- o **Debug Info Paths:**

AMP

AS2

SWD

AS3

AS

Add

Remove

- o **Load Profile Frames**

- o **Save Profile Frames**

Load Profile Frames

- o **Dump Memory Report:**AMP

"AmpMemReport.txt"

- o **Exit :**

- **View menu:**

- o **Graph Tooltips:**

AMP

- o **Time Units:**

- o **Memory Units:**Memory

Memory

- **Window menu:**

Images Fonts CPU Functions ActionScript

Log

SWF Info

Renderer

Memory

Restore UI

AMP

- **Help menu:**

Scaleform.com

2

User Guide

About

AMP

- **Application Control Toolbar**

UI

0  **Wireframe**

0  **Overdraw mode**

AMP

0  **Batch profiling mode**

0  **Anti-aliasing**

Scaleform

EdgeAA

HW

FSAA

None) 3

o  **Fast-forward** Flash

SFW

FPS

o  **Restart**

- **Frame control ツールバー:**

o  **Clear**

AMP

o  **First Frame :**

o  **Previous Frame :**

o  **Pause**

o  **Next Frame**

o  **Last Frame**

- **Profiling ツールバー:**

o  **Detailed Profiling**

:

AMP

Memory

AMP Images

10

o  **Profiling Level:**

Low

Medium

High

Low

AMP

C++

Medium

AMP

Low

C++

ActionScript

High

AMP

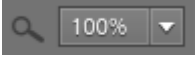
C++

High

ActionScript2

SWD

ActionScript3

-  **Zoom ツールバー:**

- **Status バー:** Status

AMP

UI
Reset UI

AMP

2 AMP の使い方

AMP
Bin/AMP GfxAmpClient.exe Scaleform

2.1 コネクションの確立

IP AMP AMP

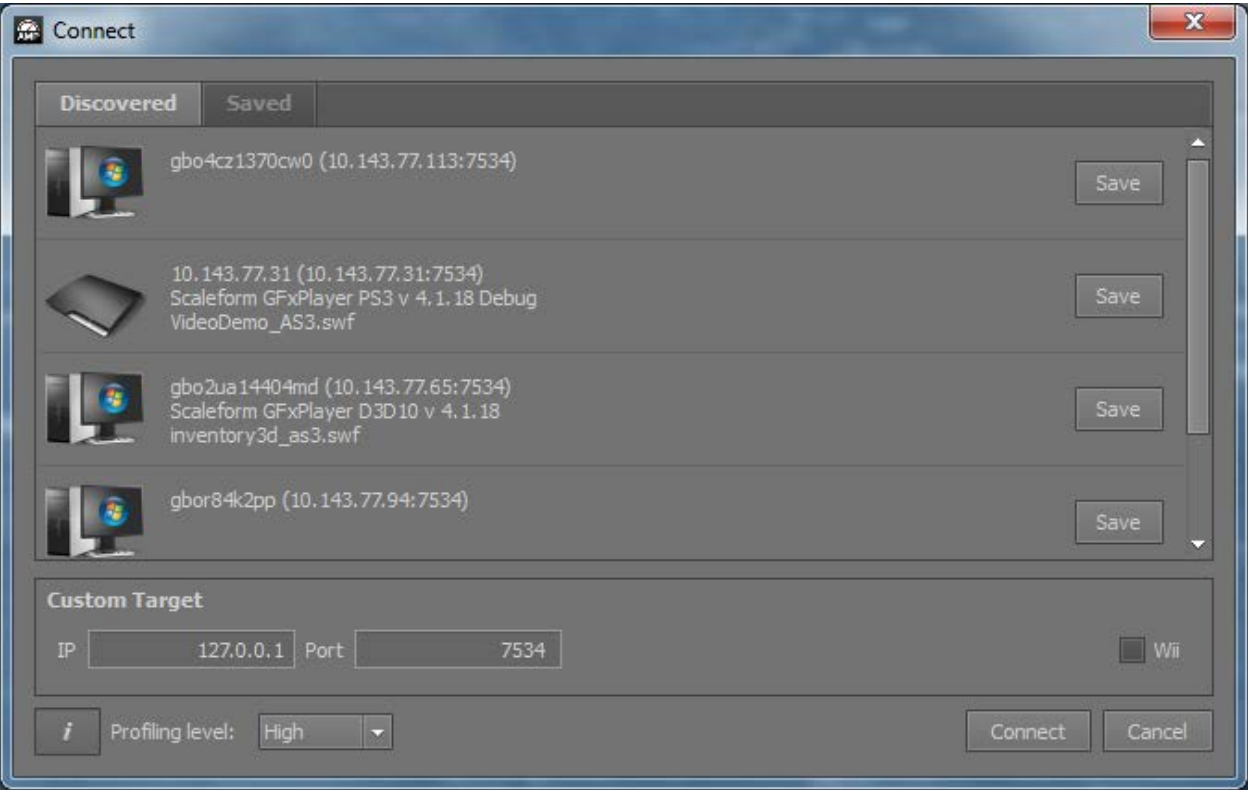
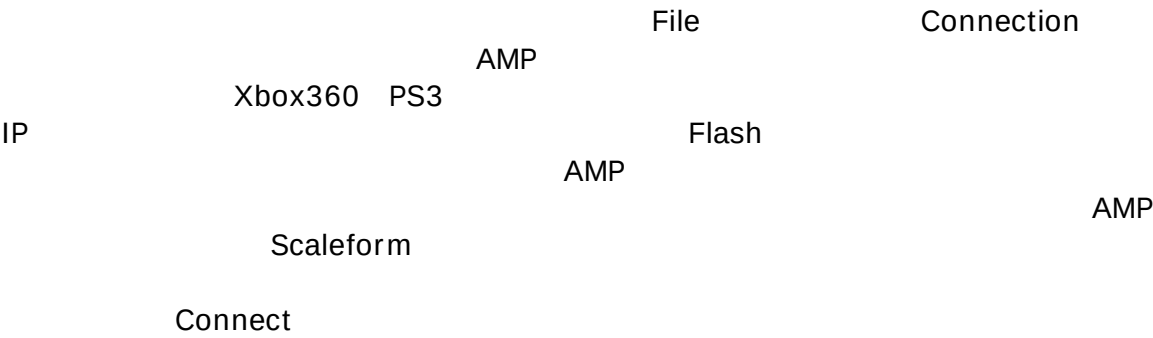


図 2 : AMP コネクション・ダイアログ



Scaleform

Wii

NDEV

NDEV

USB

Wii AMP

COM

IP

Wii

wii

2.2 性能指標の解析

Advance Scaleform Display Action Script
Direct Access API User CPU



図 3: CPU グラフ

CPU

UI

CPU

CPU Summary

CPU

ActionScript

ActionScript
Function

Log	SWF info	Renderer	Memory	Images	Fonts	CPU	Functions	ActionScript	Movie Objects
Function Name	Times Called	Time	Avg Time						
RenderThread::drawFrame	1	1.811 ms	1.811 ms						
MovieImpl::Advance - 3DGenerator_AS3.swf	1	1.106 ms	1.106 ms						
MovieRoot::DoActions - 3DGenerator_AS3.swf	1	0.017 ms	0.017 ms						
MovieRoot::DoActions - 3DGenerator_AS3.swf	1	0.770 ms	0.770 ms						
MovieRoot::ExecuteCtors - 3DGenerator_AS3.swf	1	0.001 ms	0.001 ms						
MovieRoot::ExecuteCtors - 3DGenerator_AS3.swf	1	0.001 ms	0.001 ms						
MovieRoot::ExecuteActionQueue - 3DGenerator_AS3.swf	1	0.001 ms	0.001 ms						
MovieRoot::ExecuteActionQueue - 3DGenerator_AS3.swf	1	0.620 ms	0.620 ms						
MovieRoot::ExecuteActionQueue - 3DGenerator_AS3.swf	1	0.608 ms	0.608 ms						
scaleform.clk.motion::Tween/tick	1	0.071 ms	0.071 ms						
fl.transitions.easing::Back/easeInOut	1	0.009 ms	0.009 ms						
scaleform.clk.motion::Tween/updatePosition	1	0.068 ms	0.068 ms						
scaleform.clk.motion::Tween/updatePosition	1	0.061 ms	0.061 ms						
scaleform.clk.motion::Tween/updatePosition	1	0.062 ms	0.062 ms						
scaleform.clk.motion::Tween/updatePosition	1	0.245 ms	0.245 ms						
MovieRoot::ExecuteActionQueue - 3DGenerator_AS3.swf	1	0.115 ms	0.115 ms						
MovieRoot::ExecuteActionQueue - 3DGenerator_AS3.swf	1	0.001 ms	0.001 ms						
MovieRoot::ExecuteActionQueue - 3DGenerator_AS3.swf	1	0.002 ms	0.002 ms						

図 4 : CPU Summary タブ

CPU

CPU Summary

2.3 レンダリング指標の解析

[Best Practices Guide](#)

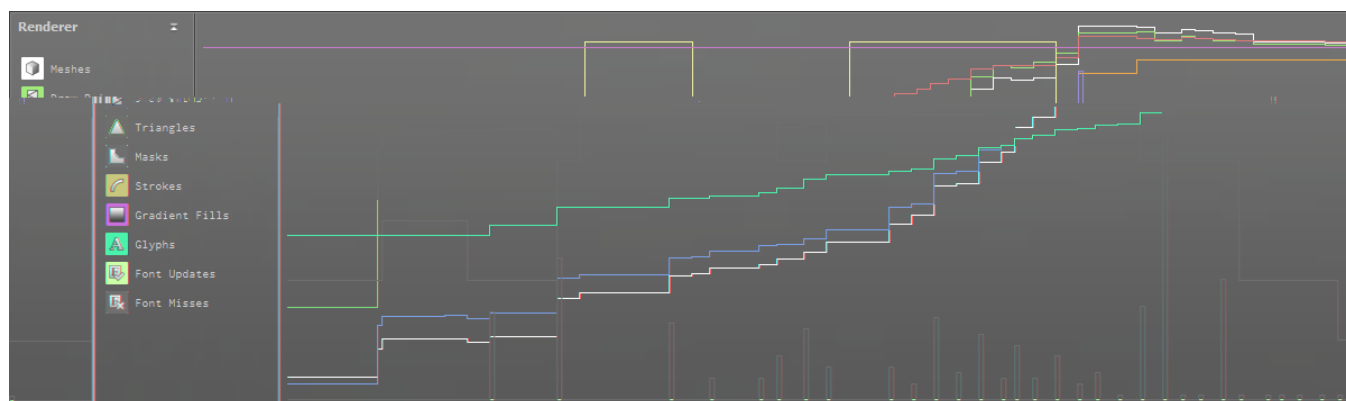


図 5: レンダリング・グラフ

Log	SWF info	Renderer	Memory
Mesher:		182	
Draw Primitives:		183	
Triangles:		5124	
Masks:		6	
Filters:		0	
Strokes:		21	
Gradient Fills:		0	
Rasterized Glyphs:		412	
Font Cache Updates:		0	
Font Textures:		1	
Font Cache Thrashing:		0	
Font Cache Fill:		28%	
Font Cache:		1048576 pixels	
Font Cache Used:		298294 pixels	
Font Failures:		0	
Font Cache Misses:		5	
Mesh Thrashing:		0	

図 6: Rendering Summary タブ

2.4 メモリ消費の解析

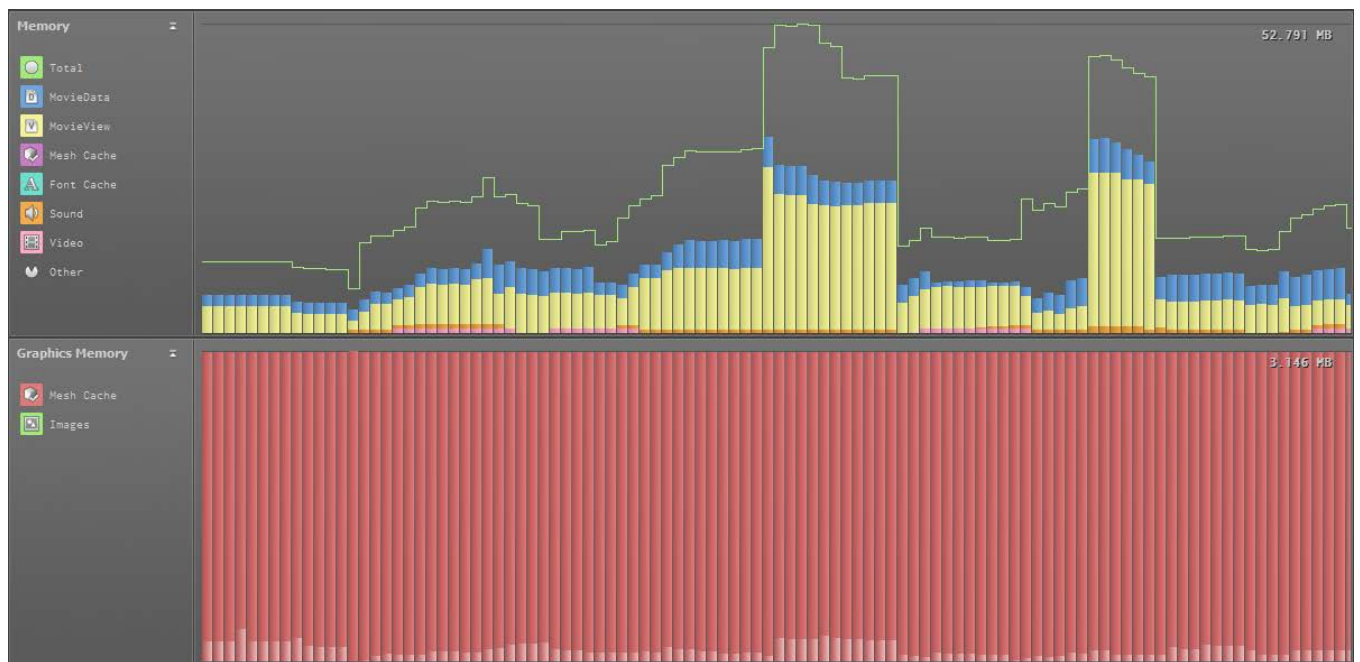


図 7: メモリ・グラフ

- **Total** Scaleform

- **MovieData** SWF GFX によって占有されて

- **Movie view** GfX::Movie

ActionScript

- **MeshCache**

- **Font Cache**

- **Sound**

- **Video**

- **Other** : Scaleform

-

Memory System Overview

“Used Space” Gfx

Debug Data Gfx
Memory

AMP i

Scaleform

Scaleform

o **Renderer:**Scaleform

- **Buffers:**
- **RenderBatch:**
- **Primitive:**
- **Fill:**
- **Mesh:**
- **MeshBatch:**
- **Context:**
- **NodeData:**
- **TreeCache:**
- **TextureManager:**
- **MatrixPool:**
- **MatrixPoolHandles:**

- **Text:**
 - html
- **Font:**
- o **MovieDef:**
 - **CharDefs:**
 - **ShapeData:**
 - **Tags:**ActionScript
 - **Fonts:**
 - **Sounds:**
 - **ASBinaryData:**ActionScript
 - **MD_Other:** MovieDef
- o **MovieView:**
 - ActionScript
 - **MovieClip:**
 - **ActionScript:**ActionScript
 - **ASString:**ActionScript
 - **Text:**
 - **XML:**XML
 - **MV_Other:** Movie View
 - **VM:**ActionScript
 - **AS3 VM:**ActionScript 3
 - **CallFrame:**ActionScript
 - **VTable:**ActionScript
 - **SlotInfo:**ActionScript
 - ClassTraits
 - InstanceTraits
 - SlotInfo
 - ClassTraits
 - InstanceTraits
 - o **SlotInfoHash:**ActionScript
 - **ClassTraits:**ActionScript
 - **Class:**ActionScript
 - **InstanceTraits:**ActionScript
 - **Instance:**ActionScript
 - **AbcFile:**
 - o **AbcConstPool:**
 - AbcFile

- **VMABCFile:**AbcFiles

- **Tracer:**

- o **IME:**Input Method Editor (IME)

Log	SWF info	Renderer	Memory	Images
Memory				Size
▲	Total Footprint			4849.402 KB
▲	Used Space			4633.715 KB
▷	Video Heaps			0.000 KB
▷	Other Heaps			325.621 KB
▲	Movie View Heaps			1226.094 KB
▲	MovieView "3dgenerator_as3.swf"			1226.094 KB
	String			0.828 KB
▷	Renderer			12.672 KB
▷	MovieView			1176.781 KB
	General			35.812 KB
▷	Movie Data Heaps			261.066 KB
▲	Global Heap			2820.934 KB
	Video			264.176 KB
	String			20.633 KB
	Sound			0.141 KB
▷	Renderer			2310.914 KB
▷	MovieView			8.039 KB
▷	MovieDef			116.676 KB
	Image			0.141 KB
	IME			9.914 KB
	General			90.301 KB
▷	Debug Memory			8.000 KB
▷	Unused Space			38.820 KB
	Debug Data			176.867 KB

8: Memory

Scaleform

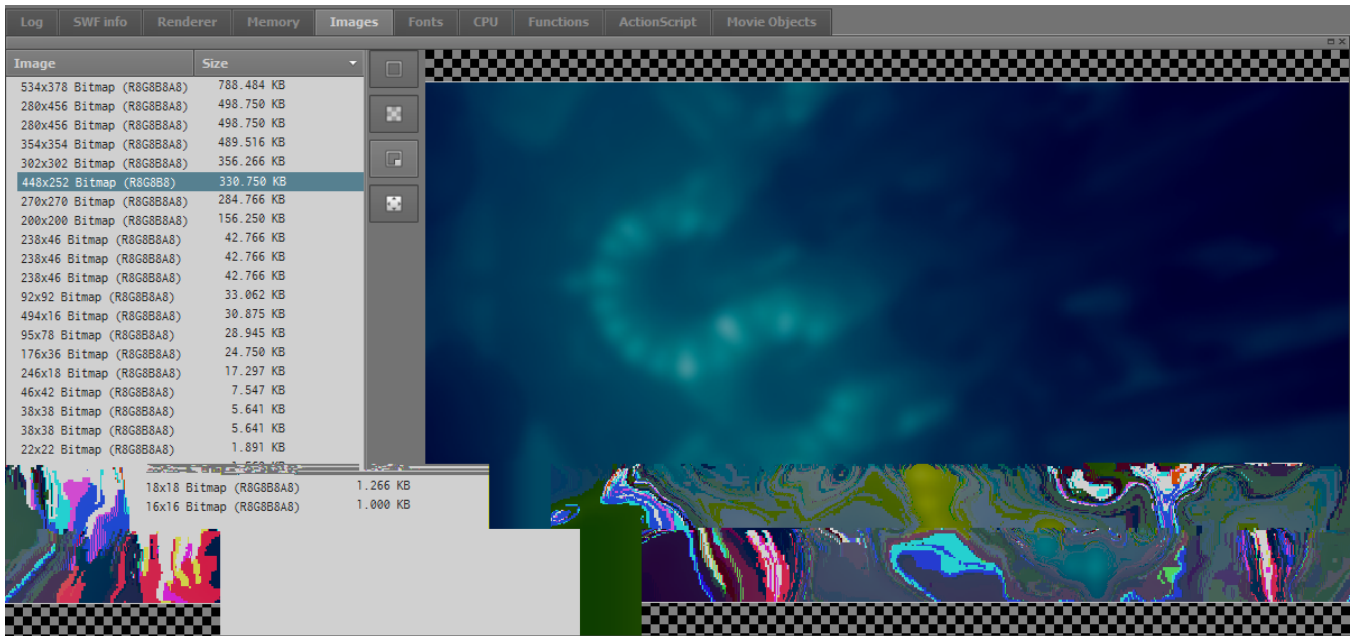


図 9: Images メモリタブ

Scaleform

Scaleform

Fonts



図 10: Fonts メモリタブ

SWF

SWF AMP MovieData CharDefs
Heaps Memory Summary

2.5 オーバードロー プロファイリングの使用

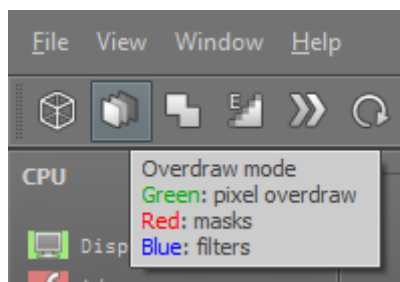


図 11: トグルボタン

16

8

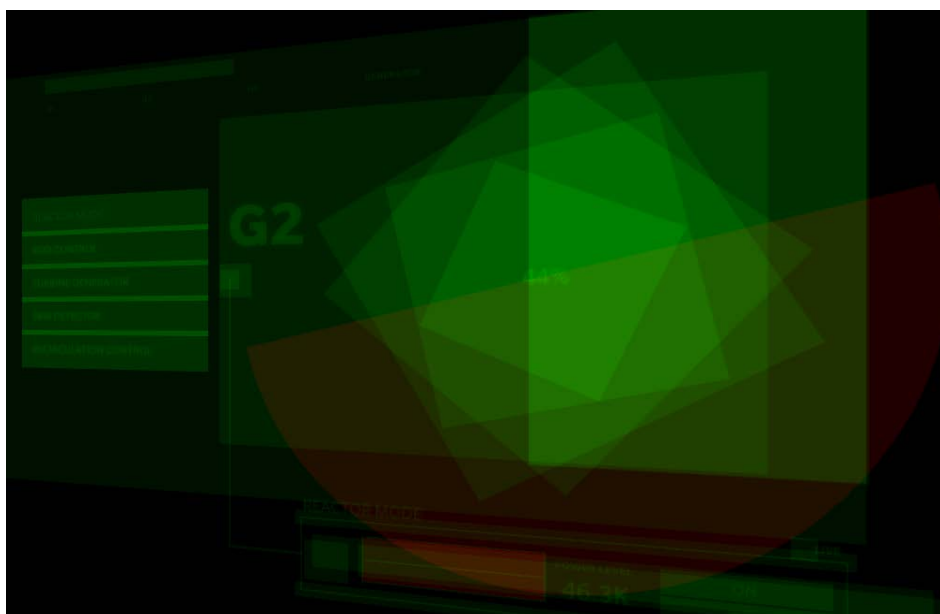


図 132: ハイライトレンダリング・モード

2.6 バッチ プロファイリングの使用

AMP

3 プラットフォームとインテグレーションに関する注意

- AMP
GFx::System
Scaleform
Scaleform::System
AMP
GFx::AMP::Server::Init() GFx::AMP::Server::Uninit()
 - AMP
iPhone/iPad, 3DS, PS Vita, WiiU
Windows Xbox360 PS3 Linux Mac Wii, Android
AMP
AMP IP
IP
 - Xbox360 IP IP
IP Xbox360 Neighborhood
 - PS3 2 IP
IP IP
 - Android
AndroidManifest.xml
`<uses-permission android:name="android.permission.INTERNET" />`
 - WiiU WiFi PC
PC 6003
TCP/IP
 - Wii NDEV USB COM Wii
NDEV hio.dll AMP IP
wii RVL_SDK/X86/bin PATH 重要 SWF AMP
SWD SWD
AMP
 - Unreal Engine 3
Flash
AS2 SWD AMP
SWF AMP
SWD
AMP

4 ビルドに関する注意

- AMP Scaleform
AMP GfxConfig.h GFX_AMP_SERVER
Scaleform Scaleform

```
AMP
Ptr<Gfx::AMP::ServerState> serverState = *SF_NEW Gfx::AMP::ServerState();
serverState->StateFlags &= Gfx::AMP::Amp_Disabled;
AmpServer::GetInstance().UpdateState(serverState);
```

- AMP

libnetctl Linux Windows Ws2_32.lib Xbox360 Xnet.lib PS3

- AMP AMP

AMP

```
AMP
AMP::Server::GetInstance().SetInitSocketLib(false)
AMP
```

- AMP Scaleform

```
AMP::Server::GetInstance().SetHeapLimit()
```

- AMP 7534

```
AMP::Server::GetInstance().SetListeningPort()
```

- AMP AMP

```
AMP::Server::GetInstance().SetConnectionWaitTime
```

5 AMP サポートの追加

Scaleform

AMP

AMP

Scaleform Player

5.1 Debug ファイルの生成

AMP ActionScript 2 Flash

SWD

Flash CS3

CS4

Ctrl+Shift+Enter

SWD

SWD

SWF

AMP

ActionScript 3

SWD

SWF

AS

Flash Studio

SWF

5.2 アプリケーションのコントロール

AMP

AMP

app-control

- AMP::AppControlInterface app-control
AMP::MessageAppControl AMP
HandleAppRequest
- AMP::Server::GetInstance().SetAppControlCallback
- AMP::Server::GetInstance().SetAppControlCaps
AMP true
AMP::MessageAppControl

```
AMP::MessageAppControl caps;
caps.SetCurveToleranceDown(true);
caps.SetCurveToleranceUp(true);
caps.SetNextFont(true);
caps.SetRestartMovie(true);
caps.SetToggleAaMode(true);
caps.SetToggleAmpRecording(true);
caps.SetToggleFastForward(true);
caps.SetToggleInstructionProfile(true);
caps.SetToggleOverdraw(true);
caps.SetToggleBatch(true);
```

```
caps.SetToggleStrokeType(true);
caps.SetTogglePause(true);
caps.SetToggleWireframe(true);
AMP::Server::GetInstance().SetAppControlCaps(&caps);
```

5.3 コネクション・ステータス

AMP

```
AMP::Server::GetInstance().SetConnectedApp
```

AMP

- **ServerStateType**
 - AMP::Server::GetInstance().SetState
 - o Amp_RenderOverdraw
 - o Amp_App_Wireframe
 - o Amp_App_Paused
 - o Amp_App_FastForward
- AMP::Server::GetInstance().SetAaMode
- AMP::Server::GetInstance().SetStrokeType
- AMP::Server::GetInstance().SetCurrentLocale
- AMP::Server::GetInstance().SetCurveTolerance
- **AMP::ServerState**
 - AMP::Server::GetInstance().UpdateState

5.4 マーカー

C++

ActionScript

CPU

ActionScript

```
Amp.addMarker(1)
```

```
Amp.addMarker() .as
```

Resources/AS2/CLIK

Amp

C++

```
Gfx::MovieImpl:: AdvanceStats->AddMarker(1)
```

6 AMP 関連の FAQ

AMP

- ## 1. AMP を動作させるのにゲーム側で最低限必要な要件は何ですか。

```

AMP                                     GFX_AMP_SERVER
      GF                               AMP
Platform::RenderThread::drawFrame                                     AMP
      AmpServer::GetInstance().AdvanceFrame

```

2. AmpClient を使用してゲームに接続しようとする、ゲームのフレームレートが非常に低くなってしまう。AmpServer::AdvanceFrame() に多くの時間が取られています。これは通常の動作ですか。

AMP

AMP

AMP

i

3. ウィンドウズ、PS3、Xbox でゲーム実行中に、ゲームがそのディスカバリーウィンドウに AMP クライアントを発見した後でこれに接続できますが、統計は全て空白です。

AMP

Platform::RenderThread::drawFrame

AmpServer::GetInstance().AdvanceFrame

AMP

- #### 4. AMP がアップデートされるときにどうして 1 秒かそれ以上ポーズするのですか (AMP::Server::WaitForAmpConnection)。

AMP

1

Scaleform m

AMP

AMP

i

AMP

profile level

1MB

GfX::AMP::Server::SetHeapLimit

5. プロファイラー使用時にはどれほどのメモリーオーバーヘッドがありますか。

AMP

1MB

AmpServer::SetHeapLimit

AMP

GF

7 追加情報

AMP

- [AMP Use Cases](#)
- [Scaleform Reference](#)
- [AMP Forum](#)